Feature matching and model fitting (Week-07 Lectures 29-33)

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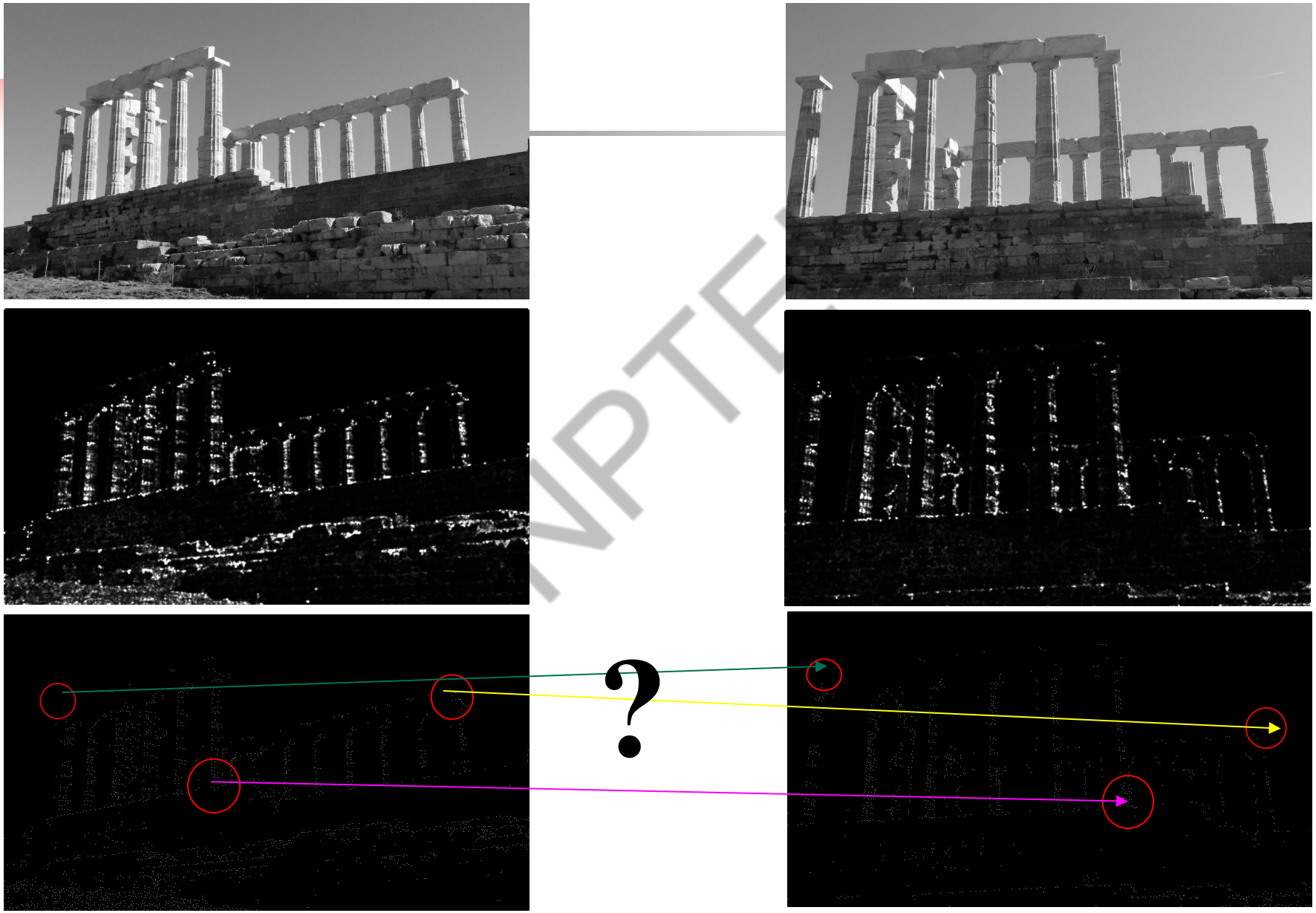
Can you get the 3-D structure of the scene?





- Get a set of pairs of corresponding points.
- Compute fundamental matrix.
- Derive camera matrices.
- Solve for 3-D coordinates of scene points for each pair of corresponding points.

Feature matching



Matching with Features

- Detect feature points in both images.
- Describe them by local statistics.
- Find corresponding pairs (Matching).

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Slide adapted from Darya Frolova, Denis Simakov, Weizmann Institute.

Matching key points

- Representation of a key-point by a feature vector.
  - e.g. $[f_0 f_1 \ldots f_n]^T$
- Use distance functions / similarity measures.
  - L_1 norm

$$L_1(\vec{f}, \vec{g}) = \sum_{i=0}^{n} |f_i - g_i|$$

  - L_2 norm

$$L_2(\vec{f}, \vec{g}) = \left(\sum_{i=0}^{n} |f_i - g_i|^2 \right)^{\frac{1}{2}}$$

  - L_p norm

$$L_p(\vec{f}, \vec{g}) = \left(\sum_{i=0}^{n} |f_i - g_i|^p \right)^{\frac{1}{p}}$$

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Weighted Distance function

- More weights to reliable components.

$$d_w(\boldsymbol{f}, \boldsymbol{g}) = \sqrt{(\boldsymbol{f} - \boldsymbol{g})^T A (\boldsymbol{f} - \boldsymbol{g})}$$

$\boldsymbol{f}, \boldsymbol{g}$: Column vectors of dimension n.

A: A +ve semidefinite matrix

- Symmetric.
- $v^T A v >= 0$ for all v.
- Typical example: Diag($w_0, w_2, \ldots w_{n-1}$), $w_i >= 0$

$$d_w(\vec{f}, \vec{g}) = \sqrt{\sum_{i=0}^{n-1} w_i (f_i - g_i)^2}$$

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A few other similarity measures

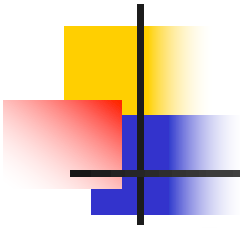
- Normalized Cross Correlation.

$$\rho(\vec{f},\vec{g}) = \frac{cov(f,g)}{s.d.(f)\times s.d.(g)}$$

$$= \frac{\frac{1}{n}\sum_{i=0}^{n-1}(f_i-\bar{f})(g_i-\bar{g})}{\sqrt{\frac{1}{n}\sum_{i=0}^{n-1}(f_i-\bar{f})^2}\sqrt{\frac{1}{n}\sum_{i=0}^{n-1}(g_i-\bar{g})^2}}$$

- Cosine similarity. $\frac{\vec{f}.\vec{g}}{\|\vec{f}\|\|\vec{g}\|}$

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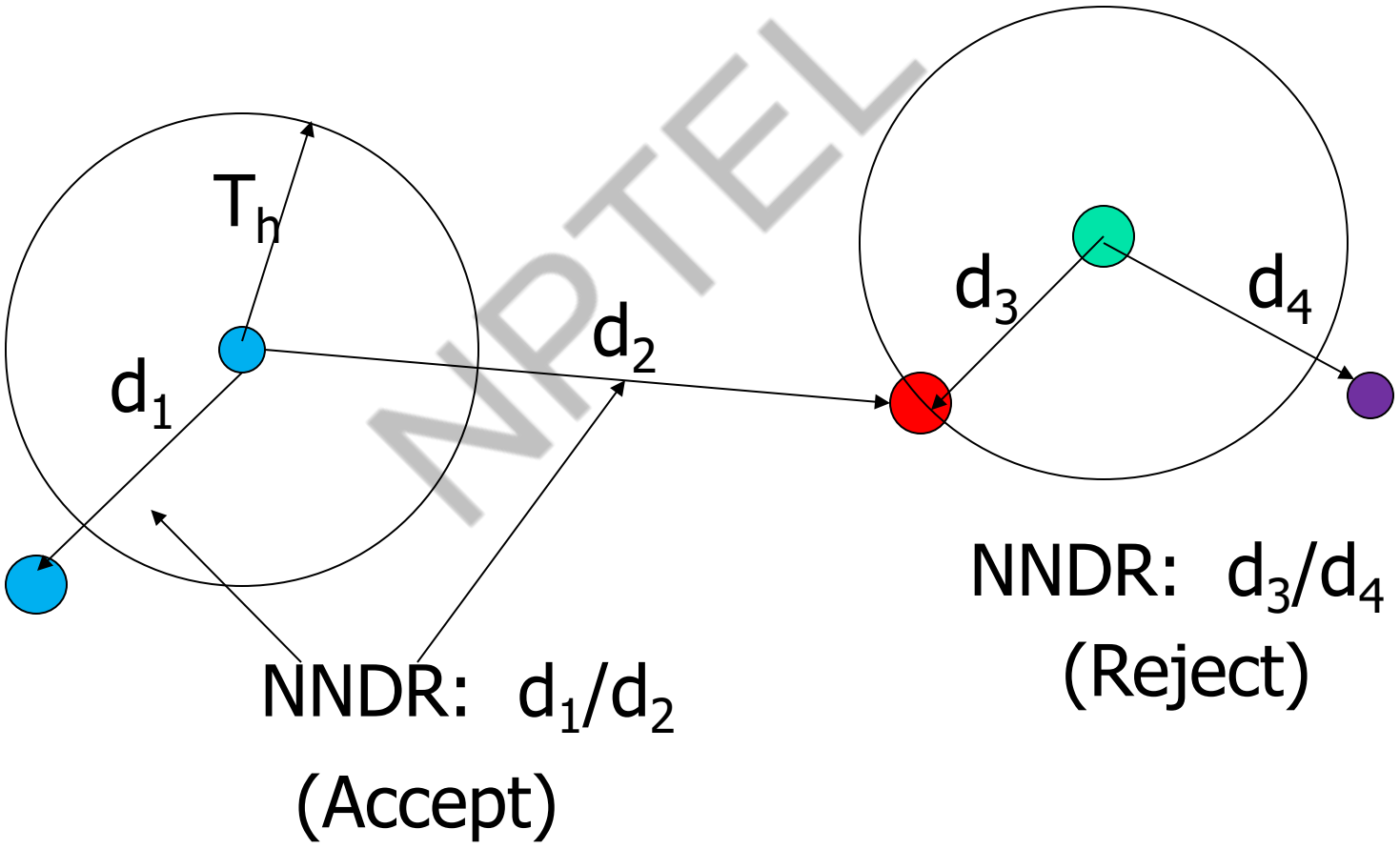


Matching criteria

- Distance based
  - Fixed threshold (FT):
    - Report all matches within the threshold value.
  - Nearest neighbor (NN):
    - Report the nearest neighbor.
  - Nearest Neighbor Distance Ratio (NNDR):
    - Report the NN if ratio of distances between the NN and 2nd NN is small.

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Matching criteria



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Matching histograms

- L_p norm of bins (representing same intervals of covariates).
  - Usually L_1 used.
- Kullback-Leiber divergence (of pdf).

$$D_{KL}(P||Q) = -\sum_{x} P(x) \ln \frac{P(x)}{Q(x)}$$

- Earth Mover's Distance (EMD)

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Earth Mover's Distance (EMD)

- EMD of two histograms P and Q measures the minimum accumulated cost of transferring masses from any bin of P to any bin of Q so that the histogram P gets transformed into Q.
  - Cost defined by the product of transferred mass and distance between bin.
    - m x $|i\text{-}j|$ for transferring mass m from ith bin of P to jth bin of Q.
    - This would make $P[i]=P[i]\text{-}m$ and $Q[j]=Q[j]+m$.
  - Total mass of P and Q should be same.
  - Accumulated cost is normalized w.r.t. to the total transfer of mass.

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H. Ling and K. Okada, An Efficient Earth Mover's Distance Algorithm for Robust Histogram Comparison, IEEE Transactions on Pattern Analysis and Machine Intelligence, 29 (5), pp. 840-853, 2007.

Computing EMD

- Two normalized histograms:
  - $P=\{p_i\}$ and $Q=\{q_i\}$, for $i=0,1,\ldots,N\text{-}1$.
  - m_{ij} transferred from i th bin of P to j th bin of Q.
  - d_{ij} distance between i th and j th bin.
- EMD is the minimum *normalized* work (transfer of masses) required for transforming P into Q.

$$EMD(P,Q) = \min_{M=\{m_{ij}\}} \left(\frac{\sum_{i,j} m_{ij} d_{ij}}{\sum_{i,j} m_{ij}} \right) \quad m_{ij} \geq 0$$

$$\sum_j m_{ij} \leq p_i \qquad \sum_i m_{ij} \leq q_j \qquad \sum_i \sum_j m_{ij} \leq min\left(\sum_i p_i , \sum_j q_j \right)$$

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H. Ling and K. Okada, An Efficient Earth Mover's Distance Algorithm for Robust Histogram Comparison, IEEE Transactions on Pattern Analysis and Machine Intelligence, 29 (5), pp. 840-853, 2007.

Computing EMD

- EMD is the minimum *normalized* work (transfer of masses) required for transforming P into Q.
- Before initiating transfer operation, bins of Q is intialized to zero. The input specificaiton q_j's are used for specifying constraints of the optimization equations.

$$EMD(P,Q) = \min_{M=\{m_{ij}\}} \left(\frac{\sum_{i,j} m_{ij} d_{ij}}{\sum_{i,j} m_{ij}} \right) \quad m_{ij} \geq 0$$

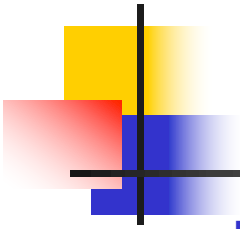
$$\sum_j m_{ij} \leq p_i \qquad \sum_i m_{ij} \leq q_j \qquad \sum_i \sum_j m_{ij} \leq min\left(\sum_i p_i , \sum_j q_j \right)$$

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Efficient computation of feature matching

- Range query
  - Nearest Neighbor of the query.
  - Report all the features within a distance from the query.
  - Ordinarily of linear time complexity O(N),
    - N is the number of features in the target image / database.
- Use of indexing and hashing.
  - K-D Tree
  - Locality Sensitive Hashing.

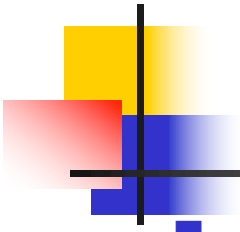
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K-D tree

- A binary tree.
- Each node contains a (key) feature vector.
- Each node behaves like a node of binary search tree (BST) for values of a particular dimension
  - Called cut-dimension.
  - Alternate periodically on nodes along any path.
  - Places a key feature vector following the rules of BST comparing on the value of the cut-dimension.

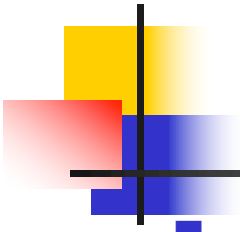
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NN Search using K-D tree

- As you walk through the tree nodes
  - Store the current smallest distance and the respective key-feature.
  - Prune sub-trees by comparing the minimum distance with the corner nodes of the hypervolume (bounding boxes) represented by the sub-tree.
  - Search the sub-tree to maximize the chance of pruning.
    - Go to the sub-tree closer to the query.

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NN Search using K-D tree

- May require to traverse all the nodes of the tree.
  - Worst case: O(N)
  - In practice close to: $O(\log(N)+2^d)$, where d is the dimension of the feature space.
    - log(N) to find the cell near the query point.
    - 2^d search around cells in the neighborhood.

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Locality sensitive hashing (LSH)

- Locality preserving hashing function (h(.)) w.r.t a distance function d(.).
  - Prob.(h(**x**)=h(**y**)) should be high, if d(**x**,**y**) is small,
    - else the probability should be small.

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LSH: A typical example

- Choose a random unit vector **r** of dimension n (e.g. following Normal distribution (N(0,1)) independently in each dimension).
- For any vector **x** of dimension n define a hash function as follows:

$$h_{\mathbf{r}}(\mathbf{x}) = \begin{cases} 1 & if\ \mathbf{x}.\mathbf{r} > 0 \\ 0 & otherwise \end{cases}$$

It can be shown, for any **x** and **y**

$$\Pr[h_{\mathbf{r}}(\mathbf{x}) = h_{\mathbf{r}}(\mathbf{y})] = 1 - \frac{\theta(\mathbf{x},\mathbf{y})}{\pi}$$

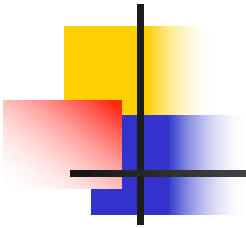
Angle between **x** and **y**.

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LSH multidimenstional bucketing

- You may also have a set of k random vectors independently generating k hash values, and define the multidimensional bucket for hashing to place the input vector.
  - $H(\mathbf{x})=[h_{\mathbf{r1}}(\mathbf{x})\ h_{\mathbf{r2}}(\mathbf{x})\ \ldots.\ h_{\mathbf{rk}}(\mathbf{x})]$
- You may have L multiple buckets (corresponding to L sets of k random vectors) for the same input.
  - $H_1(\mathbf{x}), H_2(\mathbf{x}), \ldots, H_L(\mathbf{x})$
- Given a query **y** compute L buckets and search for NN in all of them.

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Model fitting

- Given a set of data points fit a model to establish relationship among the data points.
  - Homography matrix between pixels in two images of the same planar scene.
  - Fundamental matrix between corresponding points in two stereo images.
  - Given a set of 2-D points fit a straight line / parabolic curve / circle / a high degree polynomial curve passing through them.
- Obtain data points and fit appropriate model.
  - Preprocessing, feature detection-description-matching.

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Knowledge of models

- Mathematical form of the class / family of Models.
  - 3x3 non-singular homography matrix for projective transformation.
  - 3x3 singular fundamental matrix for stereo geometry.
  - 3x4 Projection matrix.

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Choice of a model

- Error of fitting
  - Mean square error
- Likelihood
  - Pr (Data | Model)
- Size of a model
  - Number of independent parameters
- Training and Test Error
    - Large training error: Under-fitting.
    - Low training error, but large test error: Over-fitting
    - Low training and test error: A good fit.

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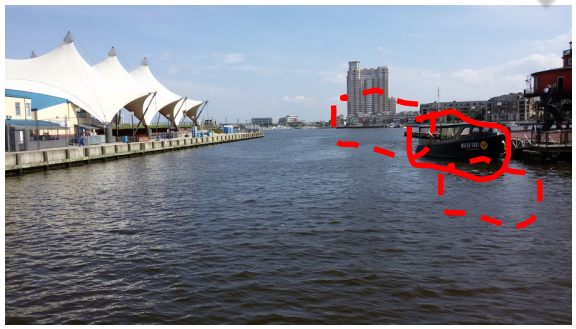
Fitting curves and lines



simple model: lines



simple model: circles

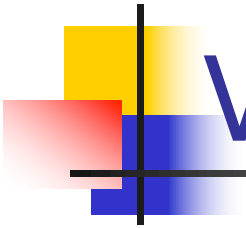


To decide about appropriate parametric models.



complicated model: Steam-boat

IPTE



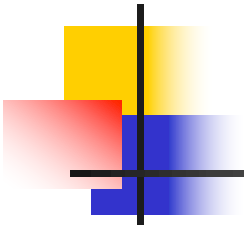
Various Issues

- **Noise / Error**
  - in estimation of feature locations.
- **Extraneous data**
  - clutter (outliers), multiple lines.
- **Missing data**
  - occlusions

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Adapted from slides by S. Lazebnik.

Fitting a straight line over 2D points



- Fitting techniques
  - Least Squares
  - Total Least Squares
  - Random Sample Consensus (RANSAC)
  - Hough Voting

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Line fitting in varying contexts

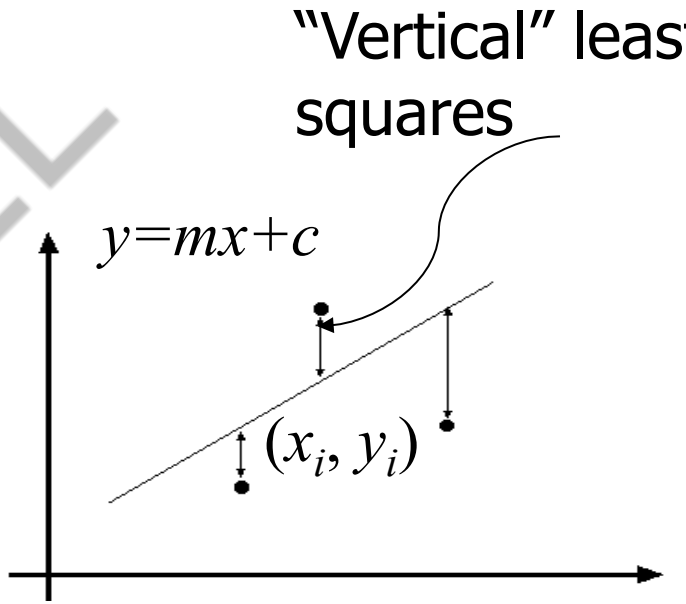
- Given points belonging to a line, to find the "optimal" line parameters.
  - Least squares
- To handle outliers.
  - RANSAC
- Presence of many lines.
  - Voting methods: Hough transform

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Adapted from slides by S. Lazebnik.

Least squares line fitting

$$E = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

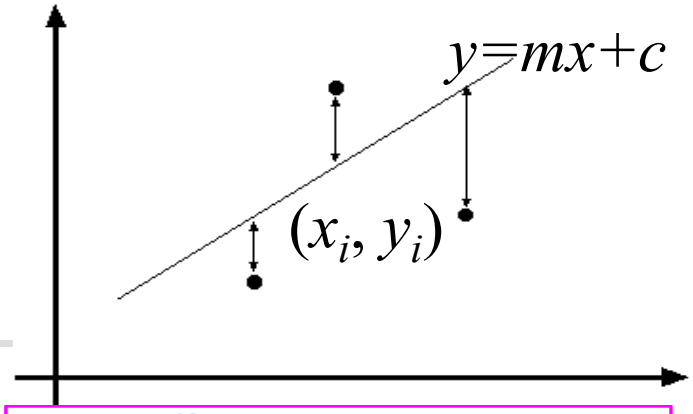


- Data:
  - $(x_1, y_1), \ldots, (x_n, y_n)$
- Line equation:
  - $y_i = m\, x_i + c$
- Find (m, c) to minimize

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Adapted from slides by S. Lazebnik.

Least squares line fitting



- Data: $(x_1, y_1), \ldots, (x_n, y_n)$
- Line equation: $y_i = m x_i + c$
- Find (m, c) to minimize

$$E = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

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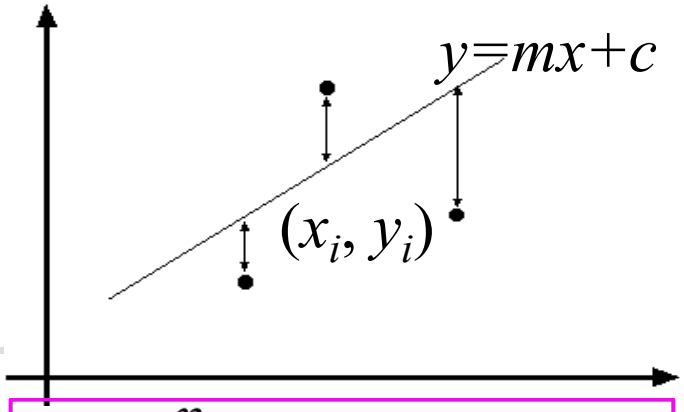
$$E = \sum_{i=1}^{n} \left(y_i - \begin{bmatrix} x_i & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} \right)^2 = \left\| \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} \right\|^2 = \|Y - XC\|^2$$

$$E = (Y - XC)^T (Y - XC) = Y^T Y - 2(XC)^T Y + (XC)^T (XC)$$

$$\frac{dE}{dC} = 2X^T XC - 2X^T Y = 0 \quad \Rightarrow \quad C = (X^T X)^{-1} X^T Y$$

Adapted from slides by S. Lazebnik.

Least squares line fitting



$$C = (X^T X)^{-1} X^T Y$$

$$E = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

$$m = \frac{cov(X,Y)}{var(X)} = \frac{\frac{1}{n}\sum_i x_i y_i - \bar{x}\bar{y}}{\frac{1}{n}\sum_i x_i^2 - \bar{x}^2}$$

$$E = n(var(y) - m^2 var(x))$$

$$c = \bar{y} - m\bar{x}$$

The R^2 goodness-of-fit criterion compares the variability in the measurements not explained by the model to the total variability in the measurements.

Goodness of fit: $R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \Rightarrow R^2 = 1 - \frac{E}{n.var(y)}$

- Not rotation-invariant.
- Fails completely for vertical lines.

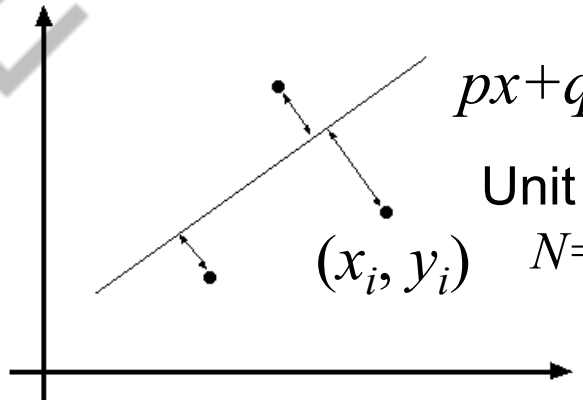
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Adapted from slides by S. Lazebnik.

Total least squares

- Distance between a point (x_i, y_i) and the line $px+qy=d$: $|px_i + qy_i - d|$ given $p^2+q^2=1$.

$$E = \sum_{i=1}^{n} (px_i + qy_i - d)^2$$



$$\frac{\partial E}{\partial d} = \sum_{i=1}^{n} -2(px_i + qy_i - d) = 0$$

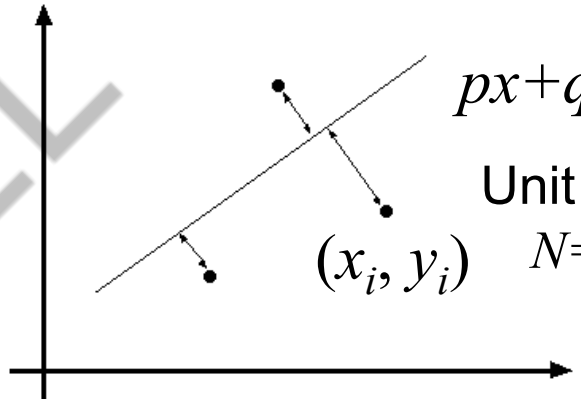
$$\Rightarrow \quad d = p\bar{x} + q\bar{y} \quad \Rightarrow \quad E = \sum_{i=1}^{n} (p(x_i - \bar{x}) + q(y_i - \bar{y}))^2$$

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Adapted from slides by S. Lazebnik.

Total least squares

$$E = \sum_{i=1}^{n} (p(x_i - \bar{x}) + q(y_i - \bar{y}))^2$$



$$E = \left\| \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} \\ \vdots & \vdots \\ x_n - \bar{x} & y_n - \bar{y} \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} \right\|^2 = (UN)^T (UN)$$

(U and N labeled above the matrix and vector)

$$\frac{dE}{dN} = 2(U^T U)N = 0$$

Subject to $\|N\|=1$.

Eigen vector of $U^T U$ corresponding to the smallest eigen value.

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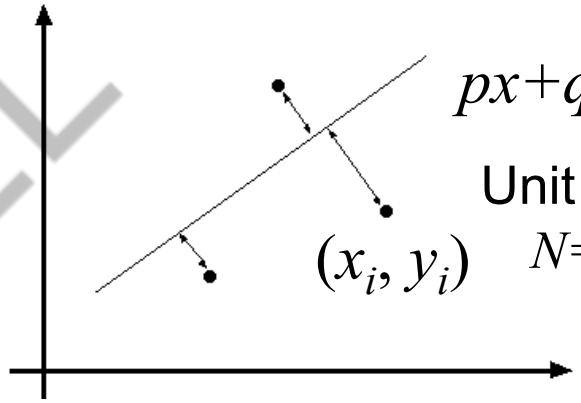
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Total least squares

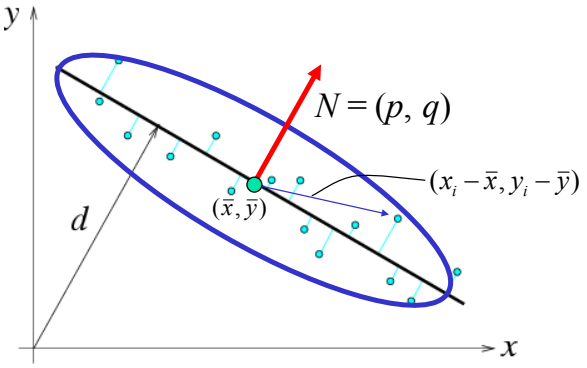
$$E = \left\| \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} \\ \vdots & \vdots \\ x_n - \bar{x} & y_n - \bar{y} \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} \right\|^2 = (UN)^T (UN)$$

$$\frac{dE}{dN} = 2(U^T U)N = 0$$

Subject to $\|N\|=1$.



$$U^T U = \begin{bmatrix} \sum_{i=1}^{n} (x_i - \bar{x})^2 & \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \\ \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) & \sum_{i=1}^{n} (y_i - \bar{y})^2 \end{bmatrix}$$



Eigen vector of $U^T U$ corresponding to the smallest eigen value.

Adapted from slides by S. Lazebnik.

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Random sample consensus (RANSAC)

- A general framework for model fitting in the presence of outliers.
- Outline
  - Choose a small subset of points uniformly at random.
  - Fit a model to that subset.
  - Find all remaining points that are "close" to the model and reject the rest as outliers.
  - Do this many times and choose the best model.

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M. A. Fischler, R. C. Bolles. Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography. Comm. of the ACM, Vol 24, pp 381-395, 1981.

Adapted from slides by S. Lazebnik.

RANSAC for line fitting

- Perform the following ***N*** times:
- Select ***s*** points uniformly at random.
- Fit a line using the LSE method.
- Find inliers to this line
  - points whose distance from the line is less than ***t***.
- For sufficient number of inliers (say > ***d***), accept the line and refit using all inliers.

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Adapted from slides by M. Pollefeys

Choice of parameters

- Initial number of points ***s***
  - Typically minimum number needed to fit the model.
- Distance threshold ***t***
  - Choose ***t*** so probability for inlier is *p* (e.g. 0.95).
  - Zero-mean Gaussian noise with std. dev. σ: t=1.96σ.
- Number of trials ***N***
  - Choose ***N*** so that, with probability *p*, at least in one trial random selections are free from any outlier. (e.g. *p*=0.99) (outlier ratio: *e*).
- Consensus set size ***d***
  - Should match expected inlier ratio: (1-e) x n.

n=No. of data points

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Estimating number of trials N

Outlier ratio: e ← Prob. that a sample is outlier.

Prob. that all s samples are inliers: $(1- e)^s$

Prob. that at least one sample is an outlier in a trial: ⇨ $(1- (1- e)^s)$

Prob. that all N trials have an outlier:$(1- (1- e)^s)^N$

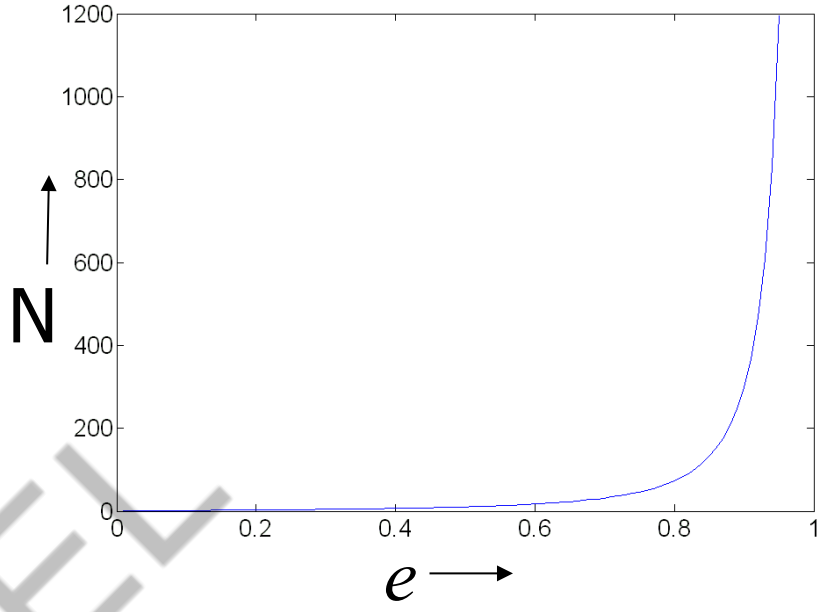
Given probability p, so that at least one random sample is free from outliers. ⇨ $(1- (1- e)^s)^N=1-p$

$$N = \frac{\log(1-p)}{\log(1-(1-e)^s)}$$

Adapted from slides by M. Pollefeys

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Choosing N



$$N = \frac{\log(1-p)}{\log(1-(1-e)^s)}$$

p=0.99

	proportion of outliers e						
s	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177

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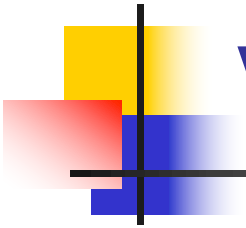
Adapted from slides by M. Pollefeys

RANSAC pros and cons

- Pros
  - Simple and general
  - Applicable to many different problems
  - Often works well in practice
- Cons
  - Many parameters to tune
  - Can’t always get a good initialization of the model based on the minimum number of samples
  - Sometimes too many iterations are required
  - Not appropriate for low inlier ratio.

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Adapted from slides by M. Pollefeys



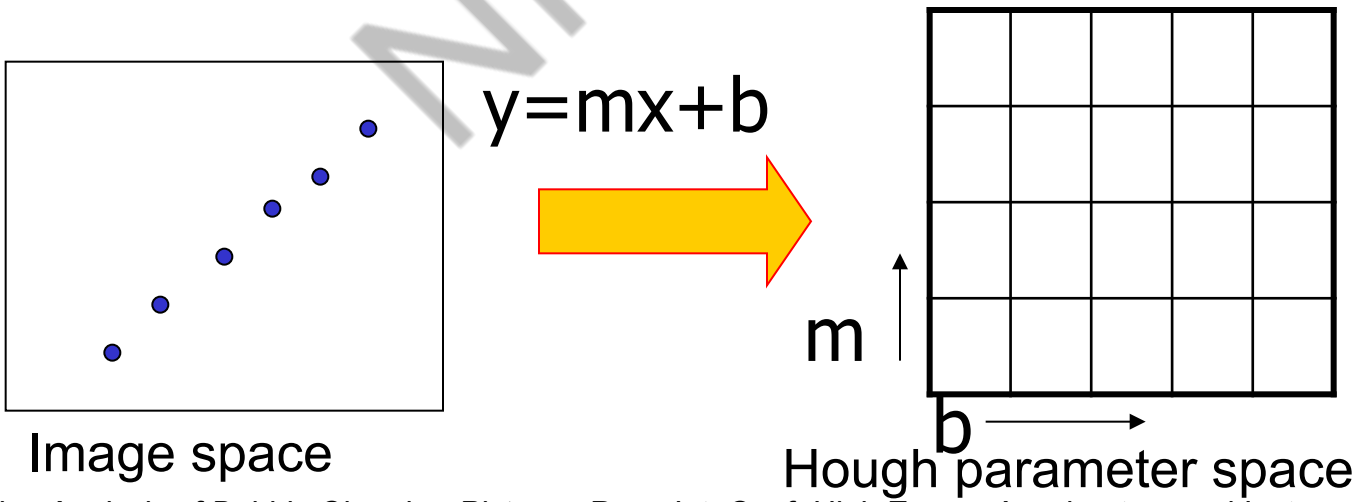
Voting schemes

- Let each feature vote for all the models that are compatible with it.
- Hopefully the noise features will not vote consistently for any single model
- Missing data doesn't matter as long as there are enough features remaining to agree on a good model.

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Hough transform

- Discretize parameter space into bins
- For each feature point in the image, put a vote in every bin in the parameter space that could have generated this point.
- Find bins that have the most votes.

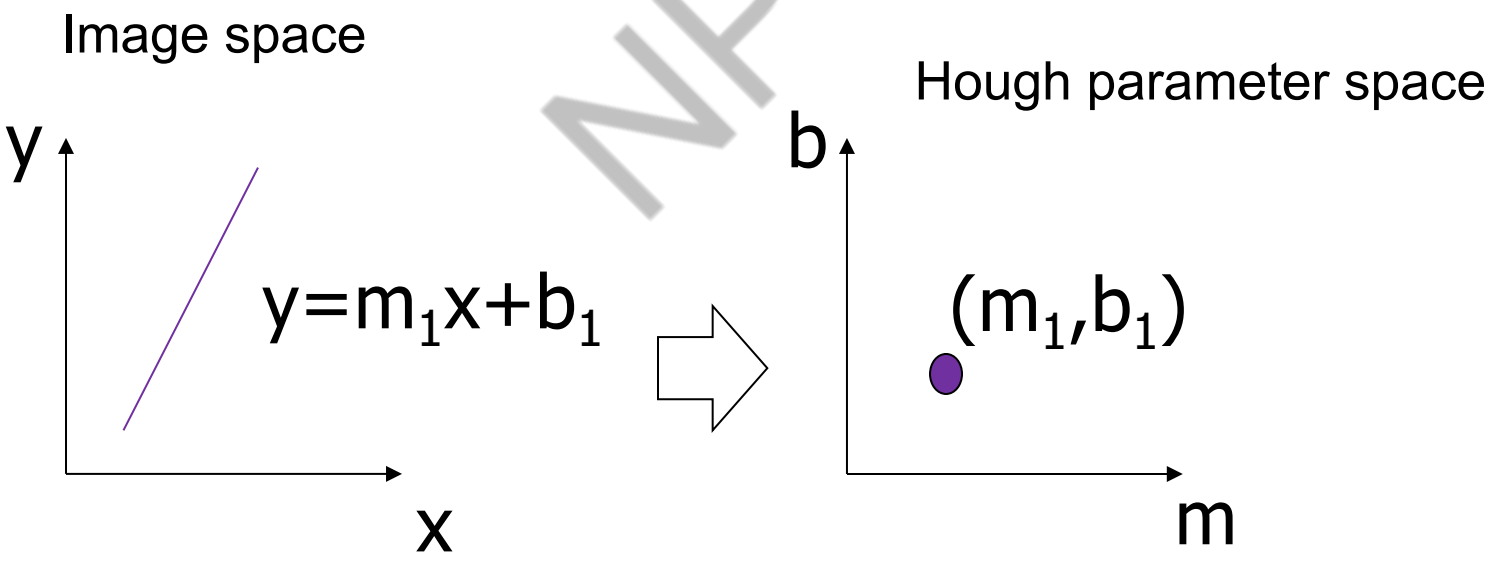


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P.V.C. Hough, *Machine Analysis of Bubble Chamber Pictures,* Proc. Int. Conf. High Energy Accelerators and Instrumentation, 1959

Parameter space representation

- A straight line in the image corresponds to a point in Hough space.



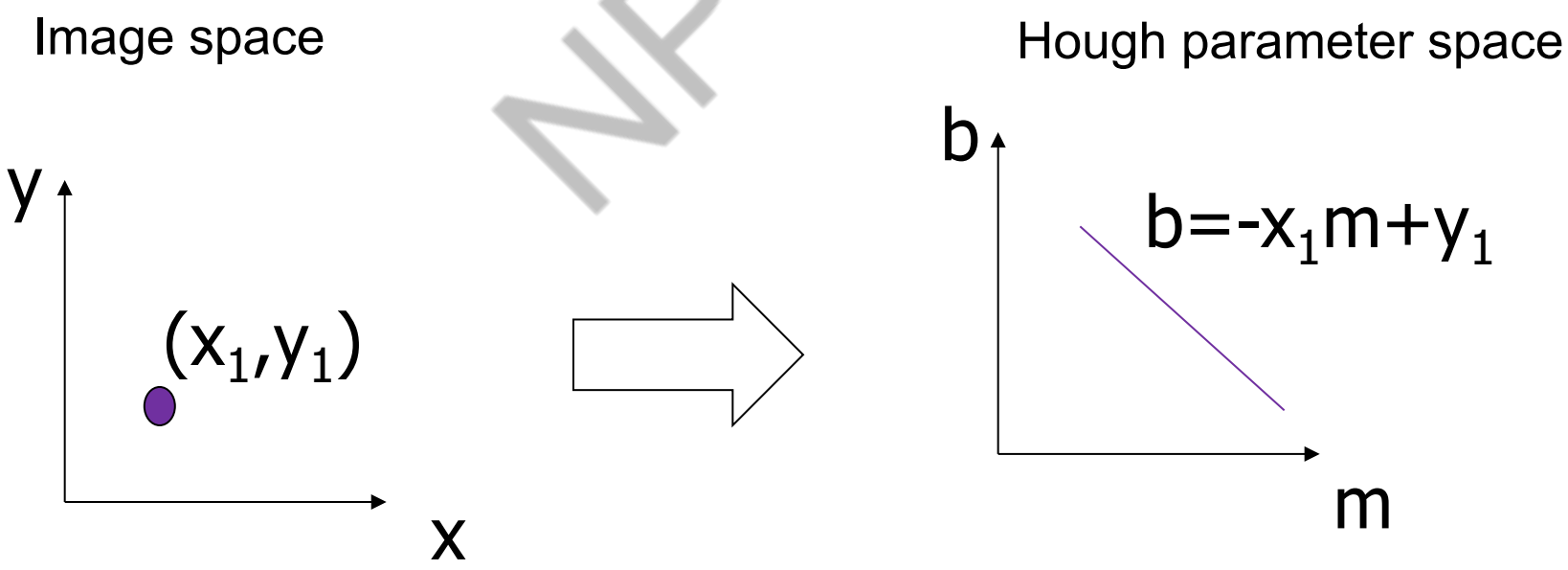
Adapted from slides by S. Seitz

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Parameter space representation

- A point in the image corresponds to a line in the Hough space.

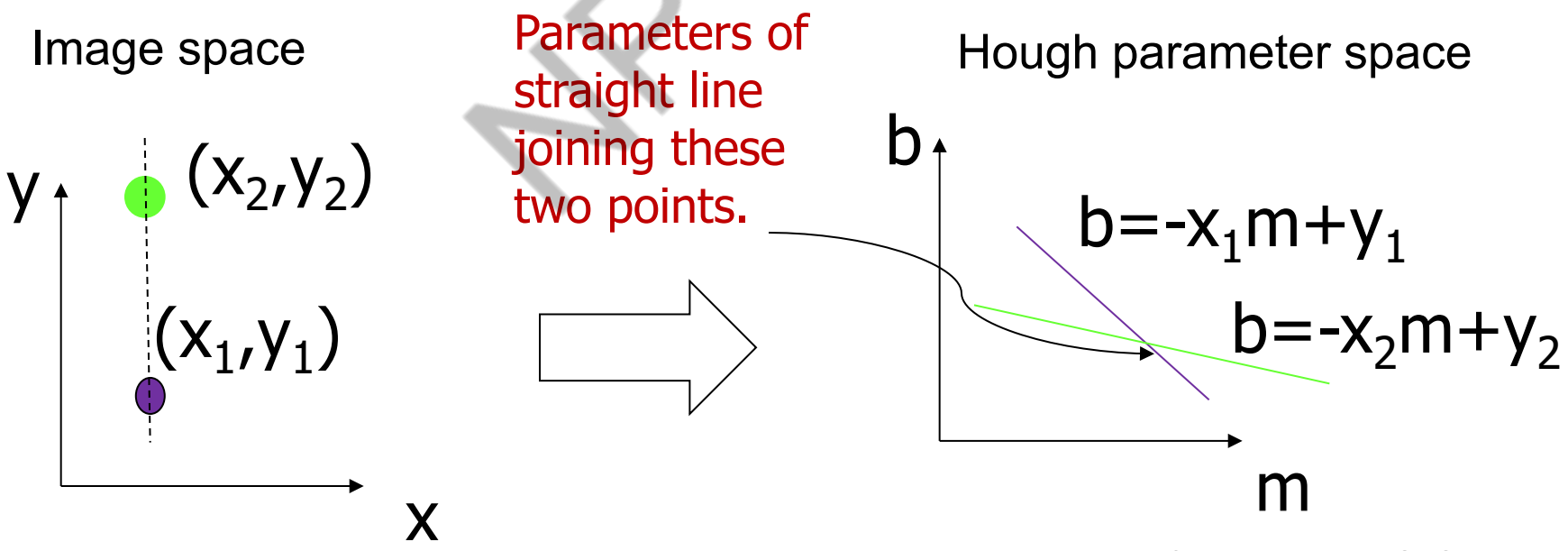
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Parameter space representation

- Two points in the image correspond to two lines in the Hough space.



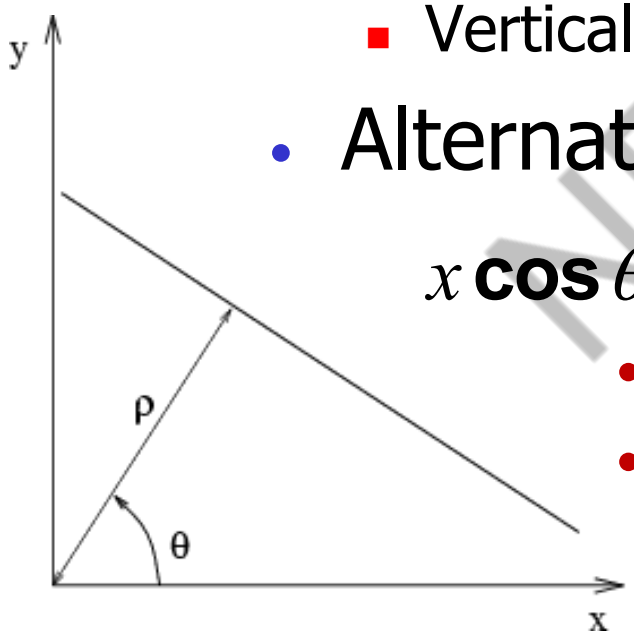
Adapted from slides by S. Seitz

Parameter space representation

- Problems with the (m,b) space:
  - Unbounded parameter domain
  - Vertical lines require infinite m
- Alternative: polar representation

$$x\cos\theta + y\sin\theta = \rho$$

  - θ varies from 0 to 180°
  - ρ varies from 0 to the length of diagonal of the image grid.



Each point will add a sinusoid in the (θ,ρ) parameter space

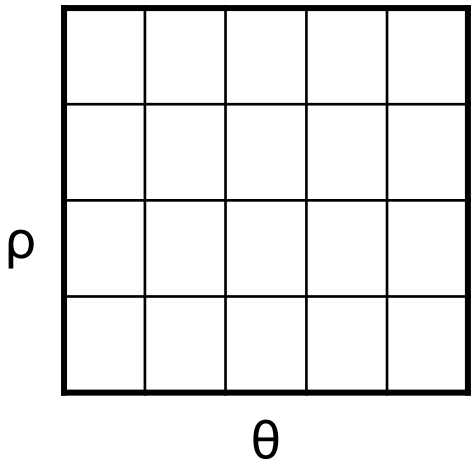
Adapted from slides by S. Seitz

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Algorithm

- Initialize accumulator A to all zeros
- For each edge point (x,y) in the image
- {
  For θ = 0 to 180
- {
  ρ = x cos θ + y sin θ
  A(θ, ρ) = A(θ, ρ) + 1
  }
  }
- Find the value(s) of (θ, ρ) where A(θ, ρ) is a local maximum
  - The detected line in the image is given by
    ρ = x cos θ + y sin θ

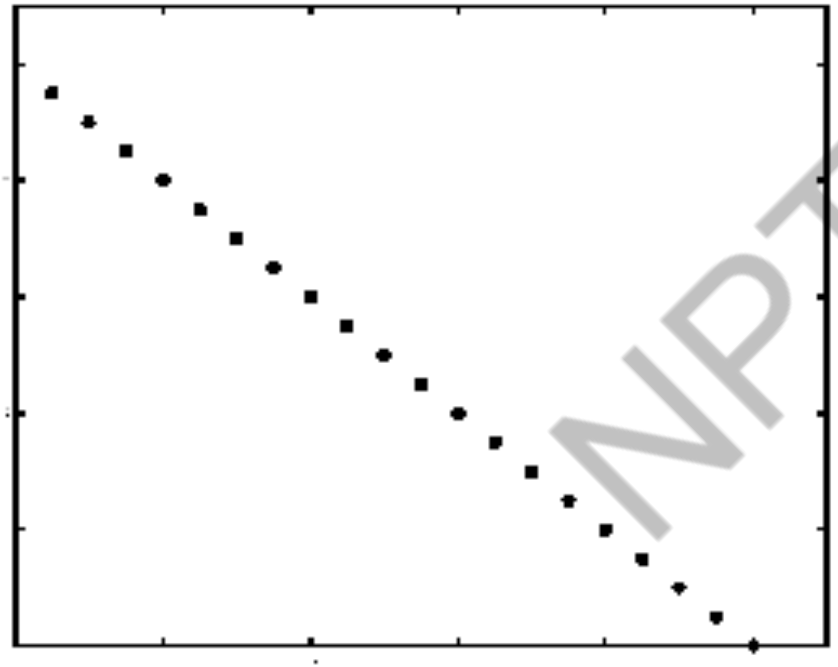
A: Accumulator array

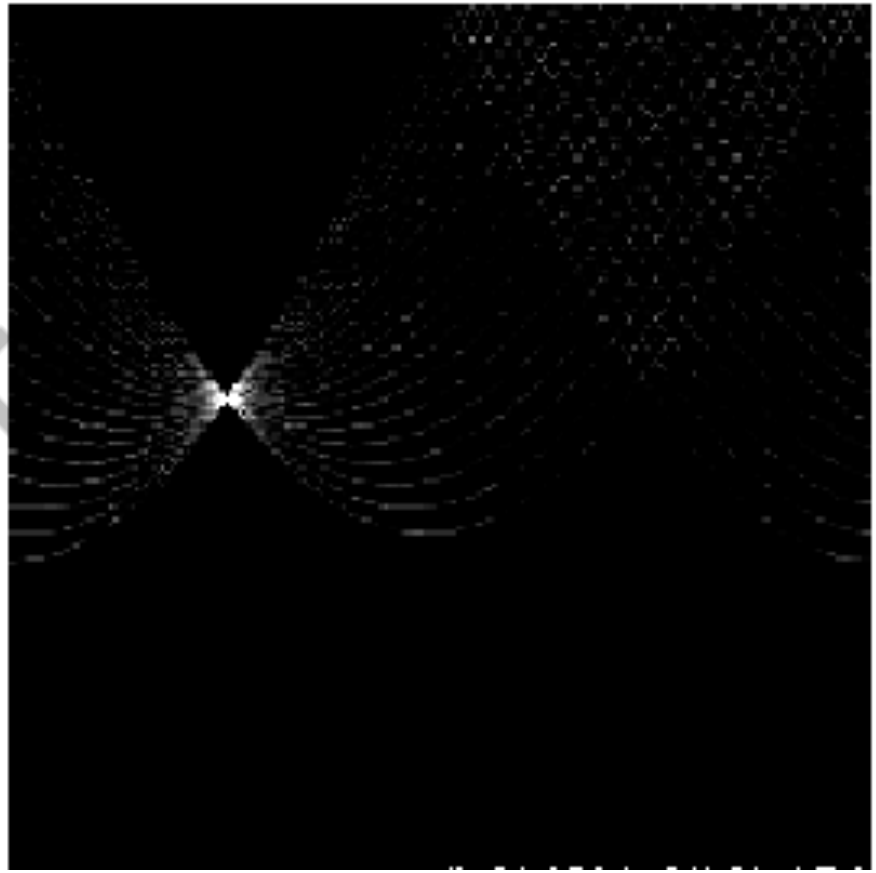


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Basic illustration





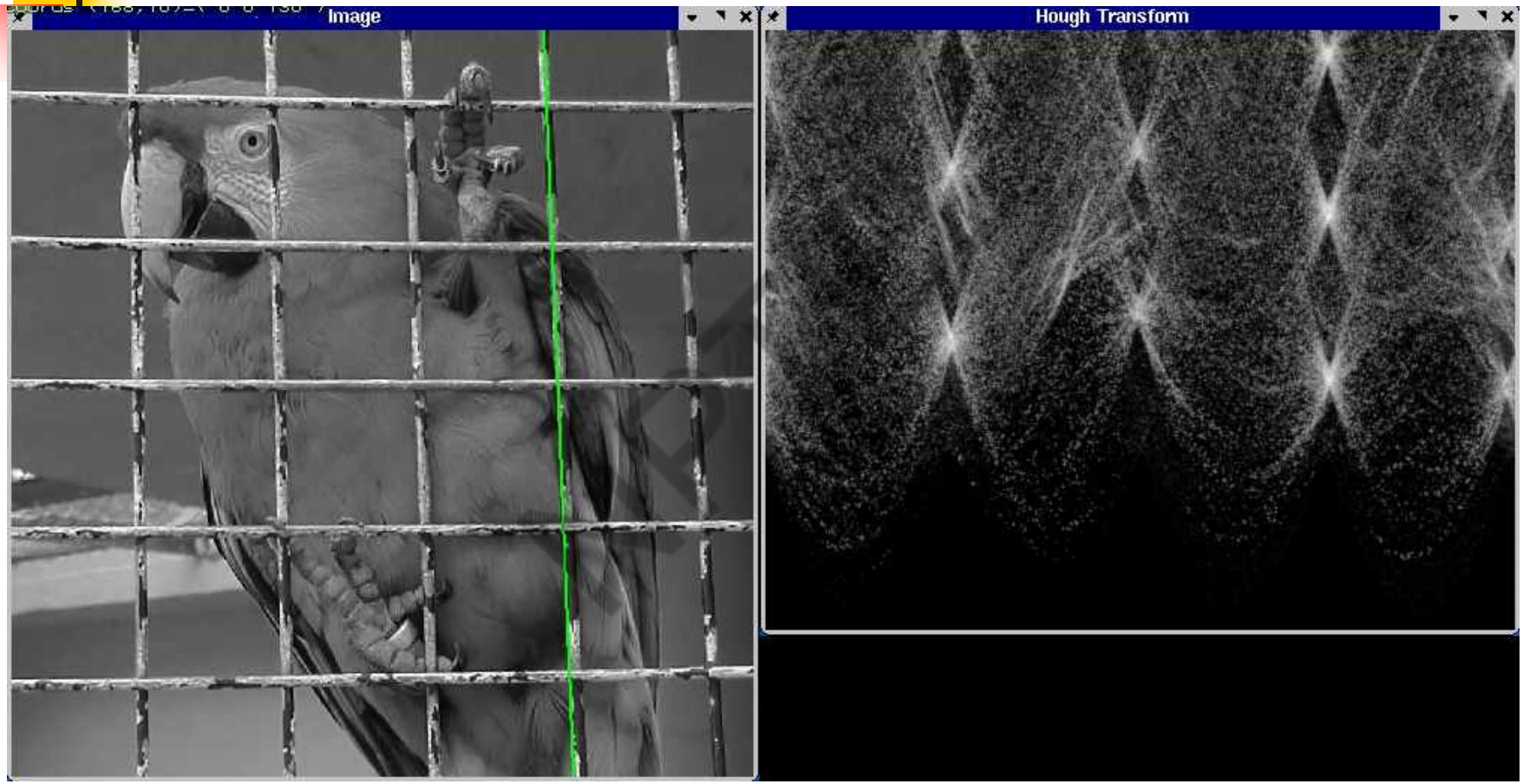
features

votes

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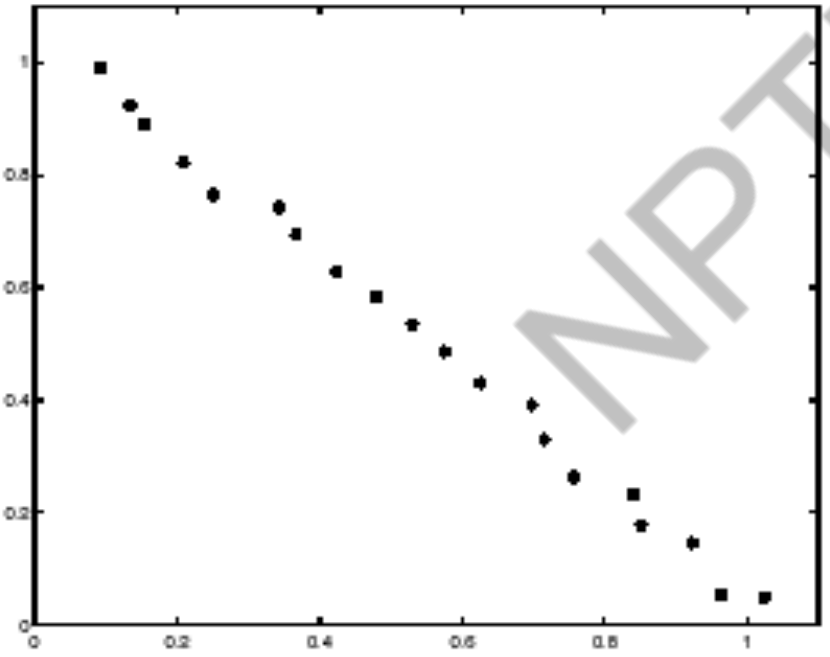
http://ostatic.com/files/images/ss_hough.jpg

A more complicated image

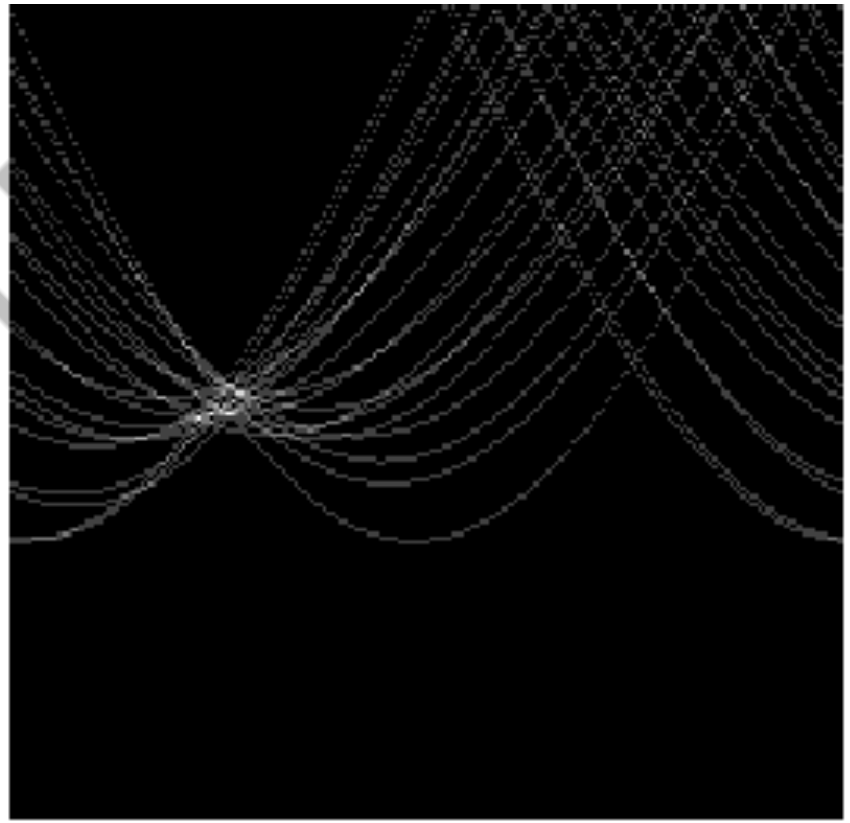


http://ostatic.com/files/images/ss_hough.jpg

Effect of noise

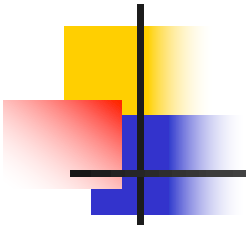


features



votes

http://ostatic.com/files/images/ss_hough.jpg



Dealing with noise

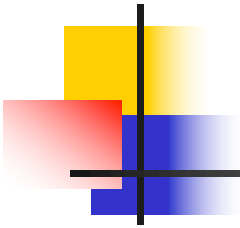
- Requires appropriate resolution of discretization of the grid in the parameter space
  - Too coarse:
    - large votes in a cell for accumulating too many different lines correspond to a single bucket
  - Too fine:
    - may miss lines as some points may not be exactly collinear voting different buckets.

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Dealing with noise

- Smoothing accumulator array.
- Try to get rid of irrelevant features
  - Take only edge points with significant gradient magnitude

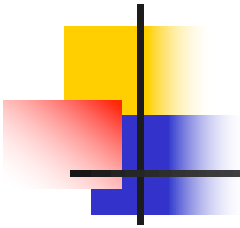
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Summary

- Feature matching.
  - Distance function / Similarity measure.
  - Different policies
    - Use of a threshold value.
    - Nearest Neighbor.
    - Nearest neighbor with distance ratio (NNDR).
  - Use of indexing / hashing for efficient computation.
  - Special distance functions for comparing histograms.
    - Kullback-Leiber Divergence (KLD)
    - Earth Mover's Distance (EMD)

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Summary

- Model fitting
  - Prior knowledge of the model useful.
  - Need to consider goodness of fit.
  - Different techniques for line fitting
    - Least squares
    - Total least squares.
    - RANSAC
    - Hough Transform.

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